GLOBAL INDUSTRIAL ARCHITECTURE



TIWAG POWER STATION
CONTROL CENTRE
(AUSTRIA)



NATIONAL ARCHIVES
PRESERVATION FACILITY
(AUSTRALIA)



BREWERY
HALL
(GERMANY)



BERTOLOTTI VIAL MACHINERY AND OFFICE
WAREHOUSE
(ARGENTINA)

GROUP 2:
NOOR ALKINANY - MAISIE RICHTER - SAM IBRAHIM - ORNELA SULCE

TIWAG POWER STATION CONTROL CENTRE

BY: NOOR ALKINANY

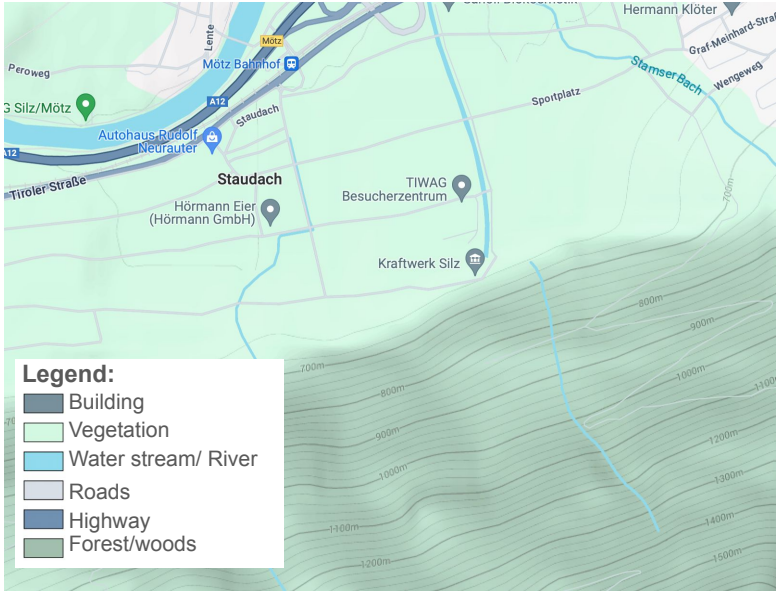




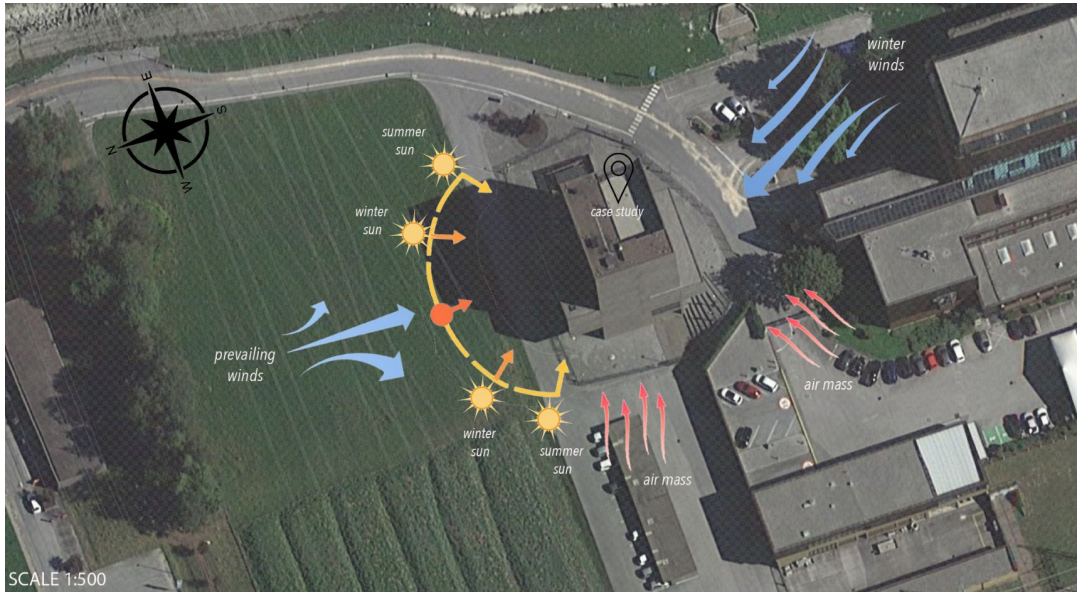
SITE ANALYSIS

Location: Silz, Austria
Address: Doktor Meinrad Praxmarer Straße, 6424 Silz, Austria
Architects: Bechter Zaffignani Architekten
Building type: Industrial (Energy Plant)
Year: 2014

Since Silz, Austria, is located in the Tyrol region, it is characterised by its mountainous terrain, forests, water streams, and alpine meadows. In summer, the sun follows a higher trajectory, providing longer daylight hours and increased solar intensity while in winter, the sun's lower path results in shorter daylight hours and less intense solar use. Furthermore, the surrounding forests, woods and vegetation help maintain the site's air quality and consistent prevailing winds.

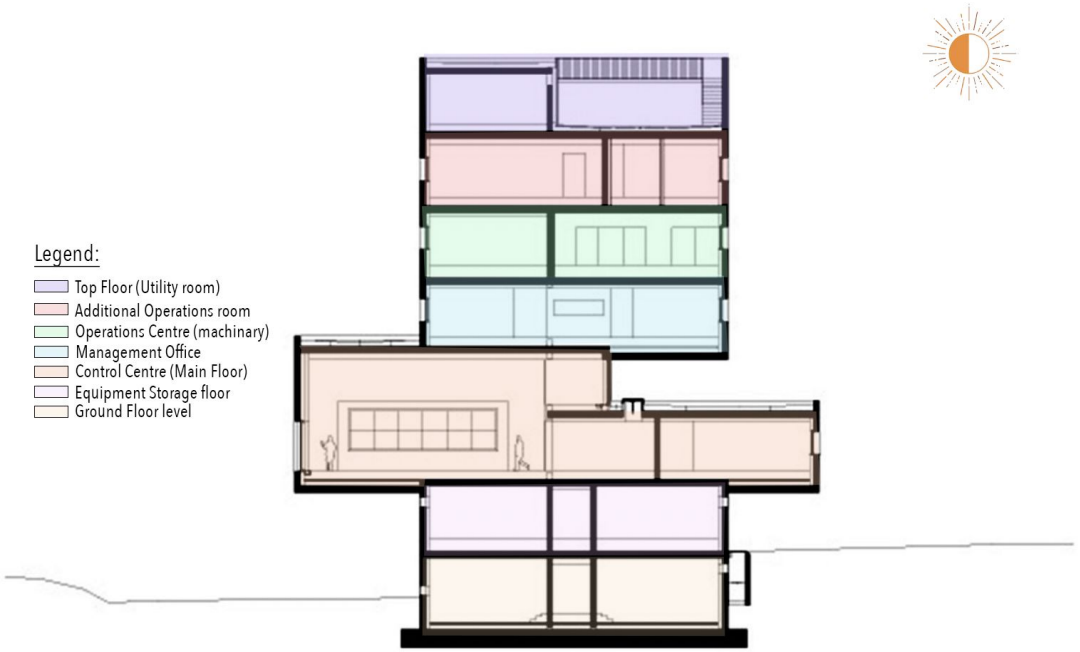


Terrain Map View

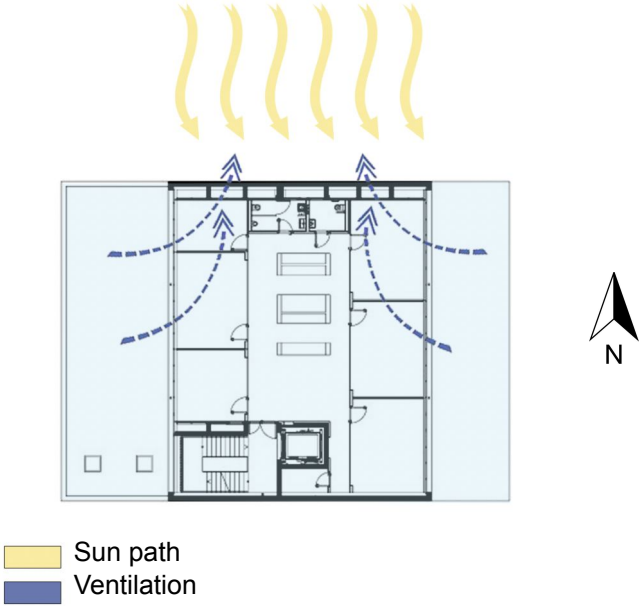


Google Earth View

FLOOR PLAN ANALYSIS DIAGRAMS

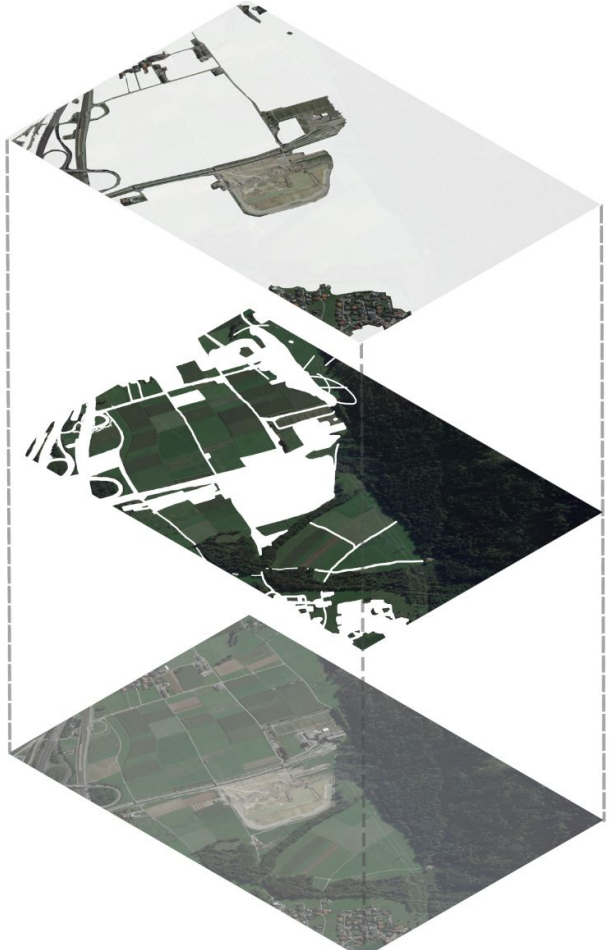


West Section Plan View



Management Office (Level 4)

3D SITE ANALYSIS DIAGRAM



Urban

Vegetation

Aerial View

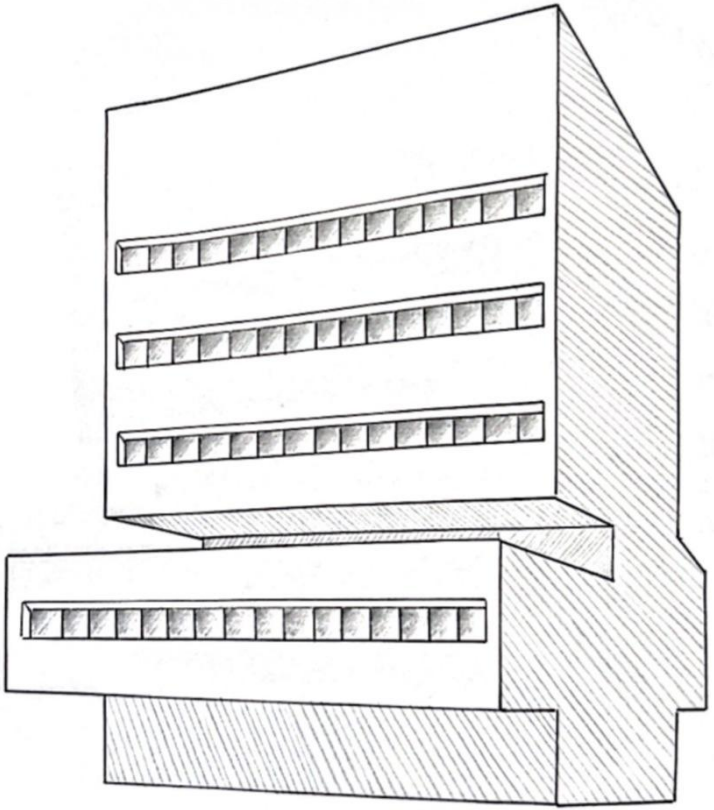
URBAN
the analysis of urban characteristics, including built environment features and human activities within and around the site.

VEGETATION
Assesses the existing vegetation cover and potential landscaping opportunities within a site.

AERIAL
Identifies the site boundary, existing infrastructure, and natural features that influence design and development

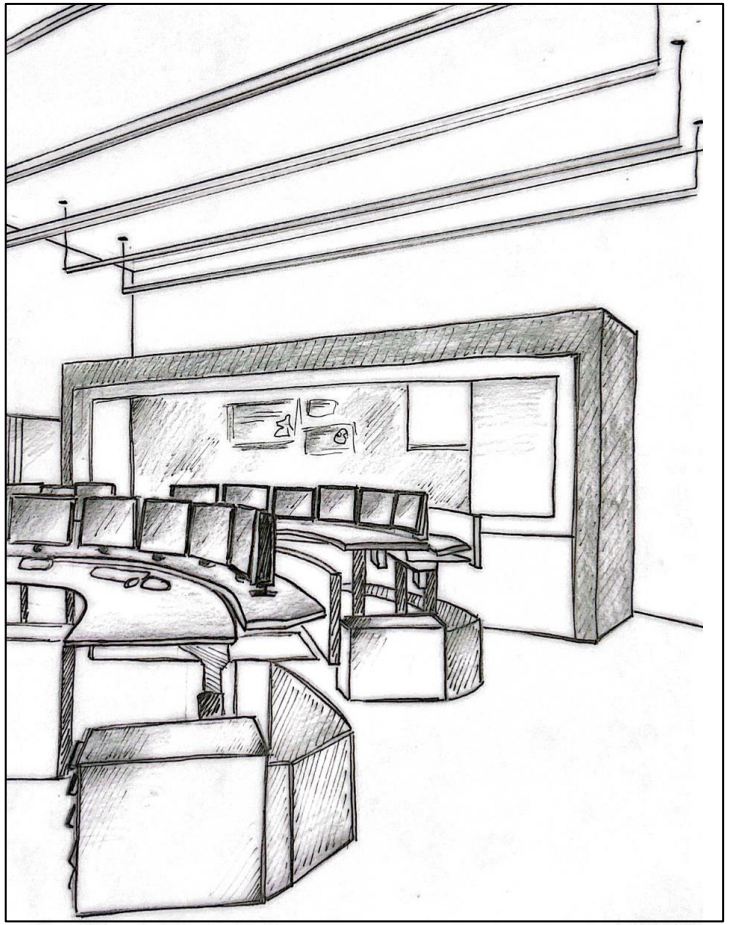
HAND-DRAWN PERSPECTIVE DRAWING 1

EXTERIOR



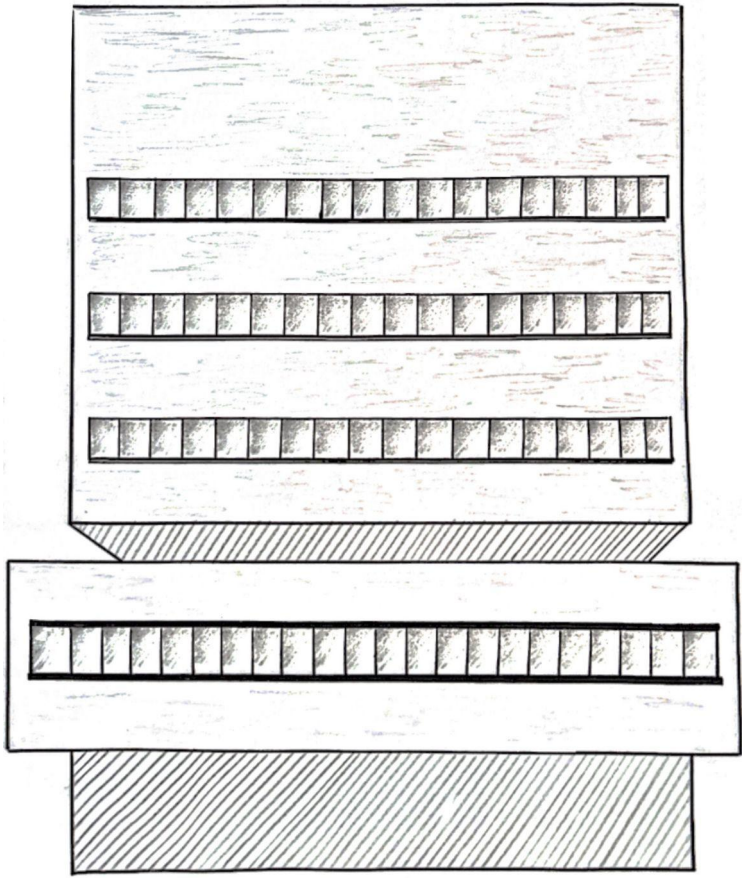
HAND-DRAWN PERSPECTIVE DRAWING 2

INTERIOR



HAND-DRAWN PERSPECTIVE DRAWING 3

EXTERIOR



MODEL MAKING NOTES

Materials needed:

- 1 x white foam board sheet (5mm)
- 1 x grey foam board (5mm)
- 1 x clear plastic sheet A2 (0.8mm)
- Hot glue gun
- PVA craft glue
- Scale ruler
- Box cutter/Stanley knife
- Cutting board
- Duct tape
- Paint (grey)

Plan:

- Purchase all needed materials
- Print out case study photos of all sides and angles of building
- Follow shapes and structures of model and replicate on foam board
- Use box knife and cutting board to cut out the shapes
- Glue together main structural parts
- Measure and cut out plastic sheets to scale
- Use PVA glue to attached sheets for glass windows
- Paint all the pieces grey to represent concrete material
- Glue everything together to create 3D model
- Allow 2-3 days for glue to completely dry
- Take professional photos and prepare for the presentation

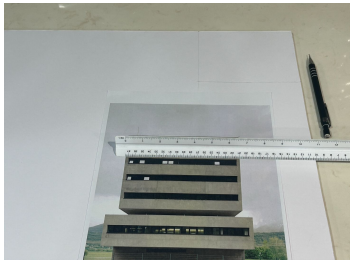
Costs:

- The overall cost for all materials is $47 (excluding things I already own)
- Materials bought: 08/04/2024

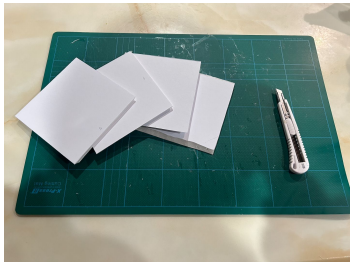
MODEL MAKING PROCESS



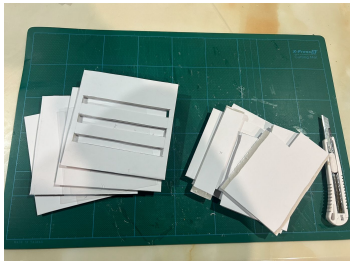
Step 1: supplies bought



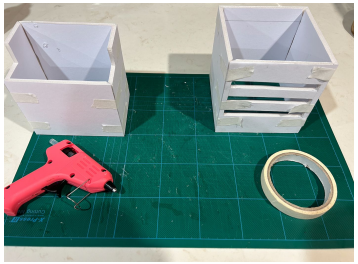
Step 2: measuring and drawing main shapes to scale



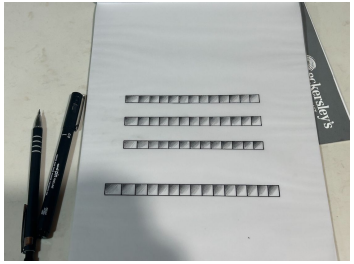
Step 3: starting to cut out main shapes



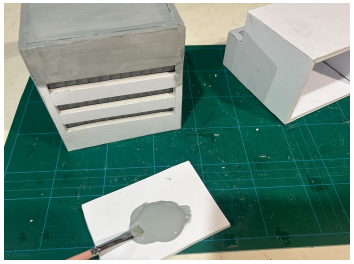
Step 4: cutting out details (windows) & other shapes



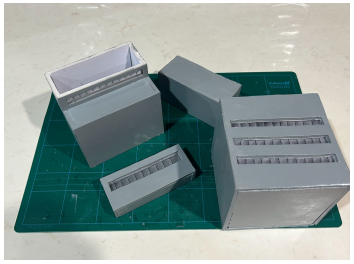
Step 5: starting to glue pieces together (using tape to hold them tight)



Step 6: drawing the window details



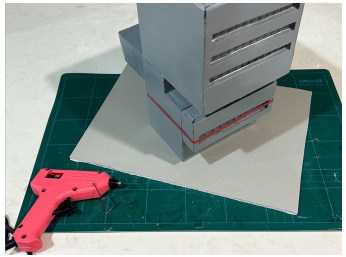
Step 7: painting all the pieces grey



Step 8: all pieces painted, windows done

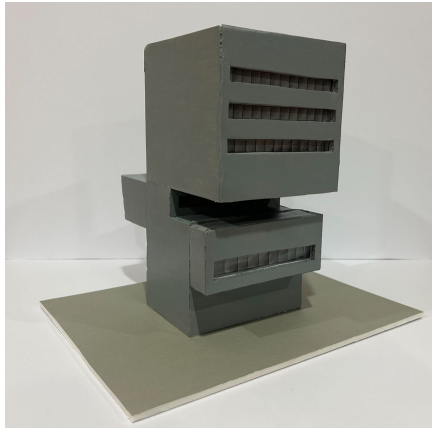


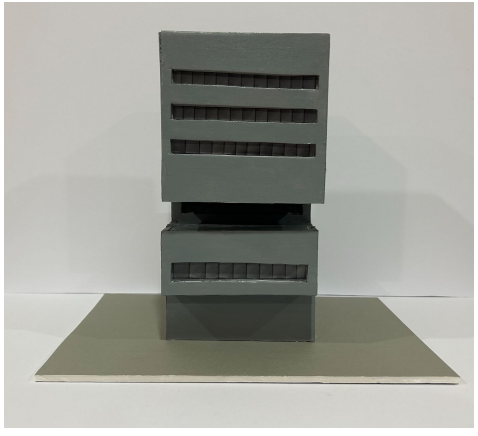
Step 9: cutting out grey foam for model base

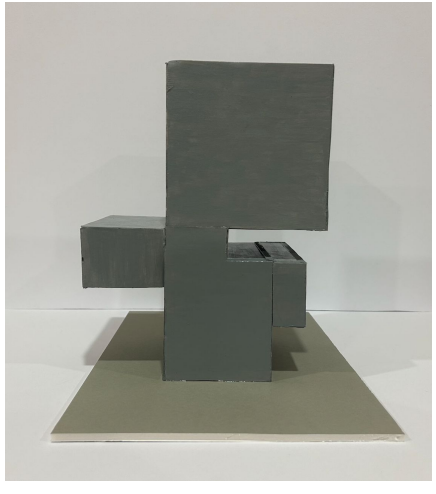


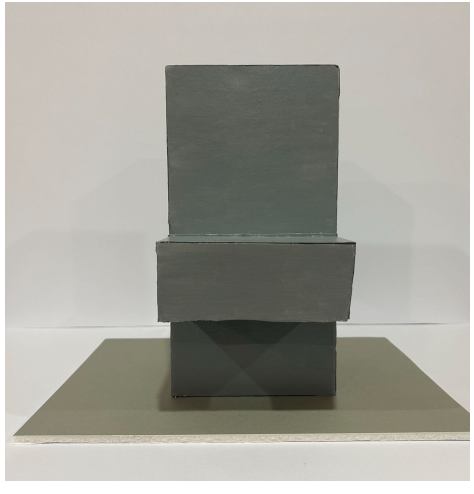
Step 10: gluing everything together

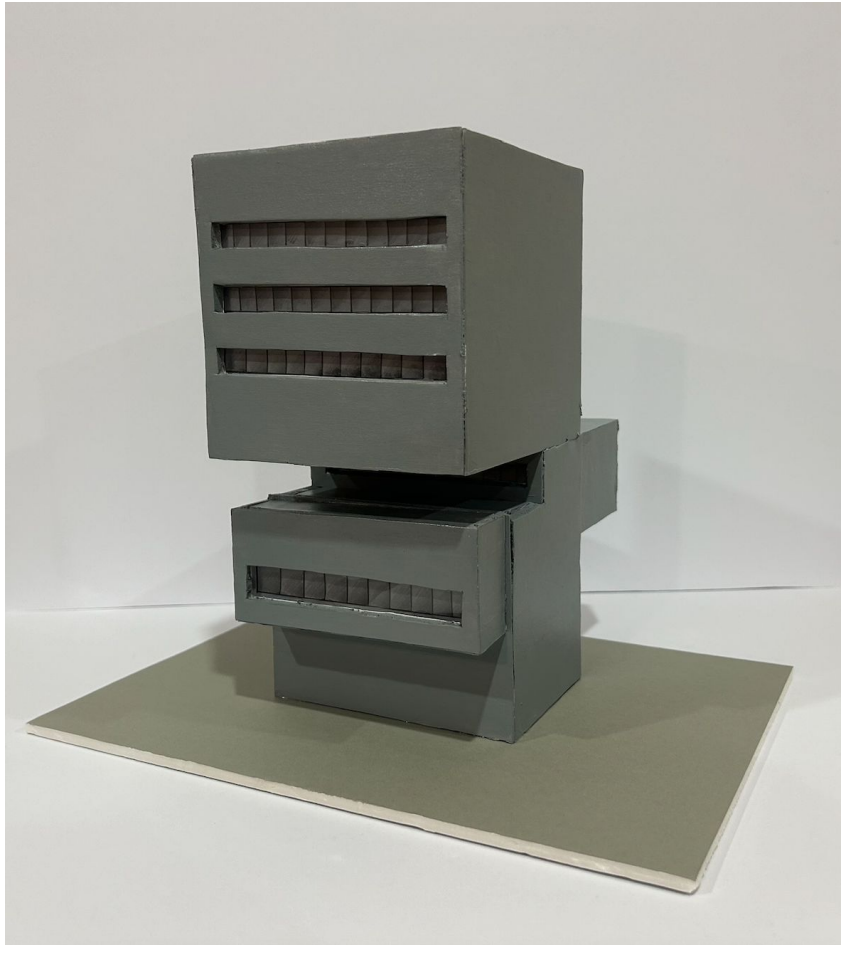
FINSIHED MODEL











REFLECTION

Group Theme Summary:
Theme: Global Industrial Architecture
Members: Noor, Ornela, Sam & Maisie (Group 2)
Colour Scheme: Black & White
Aesthetic: Minimalistic Presentation
Model Material: Foam Board

Our group has agreed to focus on case studies based on the theme of "Global Industrial Architecture" as we all shared a mutual interest in the intrinsic significance of the way the built environment is shaped and reflected. Industrial Architecture has allowed us to explore the interplay between form and function, as these chosen case studies must balance practical requirements with aesthetic considerations. It also serves as a lens through which to examine broader issues such as urbanisation, globalisation and sustainability, making it a compelling and relevant field of study which will benefit our academic studies in the future.

Through the exploration of various case studies, I have decided to focus my attention on the energy plant building in Austria 'TIWAG Power Station Control Centre'. Beyond its functional significance, I was drawn towards the the visually intriguing architectural design as I have found I am attracted to simple and minimalistic designs. I also have a deep love for nature, and since the case study is situated amidst Silz's natural scenery, I was greatly intrigued by the built environments connection with the surrounding scenery.

Concrete is the optimal choice for withstanding the powerful force of water harnessed at the TIWAG Power station as it reflects the unique character of the site. This Robust material used conveys a sense of security and durability, harmonising with the surrounding forests, woods, and vegetation that contribute to the site's air quality. The brown colour of the smooth exposed concrete surfaces, achieved with iron oxide pigment, symbolises the dynamic nature of power generation amidst this natural setting in Silz Austria.

The decision to use foam boards for my physical model reflects both practical and aesthetic considerations. Foam boards offer a clean and neutral backdrop that enhances visibility and contrasts with architectural details, allowing me to effectively showcase the building's form, materials, and layout.

This whole process has provided me with insights both into the architectural design principles and the practicable process of building a physical model. The power station's use of materials like concrete to withstand the force of water in hydroelectric generation underscores the importance of selecting durable and reliable materials that also convey a sense of longevity. I've also learned to understand the site's natural surroundings and the integration of environmental factors into architectural decision making for sustainable infrastructure.

Building the model of the case study has allowed me to learn valuable techniques and creative problem solving skills in architectural modelling. Model building translated my theoretical knowledge into a tangible representation that captures the essence of the TIWAG Power station's design and function. This project has deepened my understanding of the intricate relationship between architecture, materials and context, emphasising the importance of thoughtful design in global industrial architecture.

Global Industrial Architecture

Selected Case Study: National Archives Preservation Facility

By Maisie Richter

Group - Maisie, Noor, Sam, Ornela

Site Analysis Stage 1

Site Analysis

Typology - Ecological Function: Ecological function of vegetation on the site, such as providing habitat for wildlife, improving soil stability, enhancing biodiversity, and mitigating environmental impacts such as erosion and stormwater runoff for the facility.

Address: 30 Vicars St, Mitchell ACT 2911

Pedestrian circulation through car park.

Pedestrian /worker circulation around facility.

Sun Path and Climate:The ACT typically sees daily high temperatures in summer ranging from 26-28°C in the northern regions to approximately 18-20°C in the alpine areas. Winter brings average low temperatures ranging from 2-4°C in the north to around -4 to -2°C in the southern parts and the alpine regions. Throughout the year, this territory vast and distinct seasonal changes.

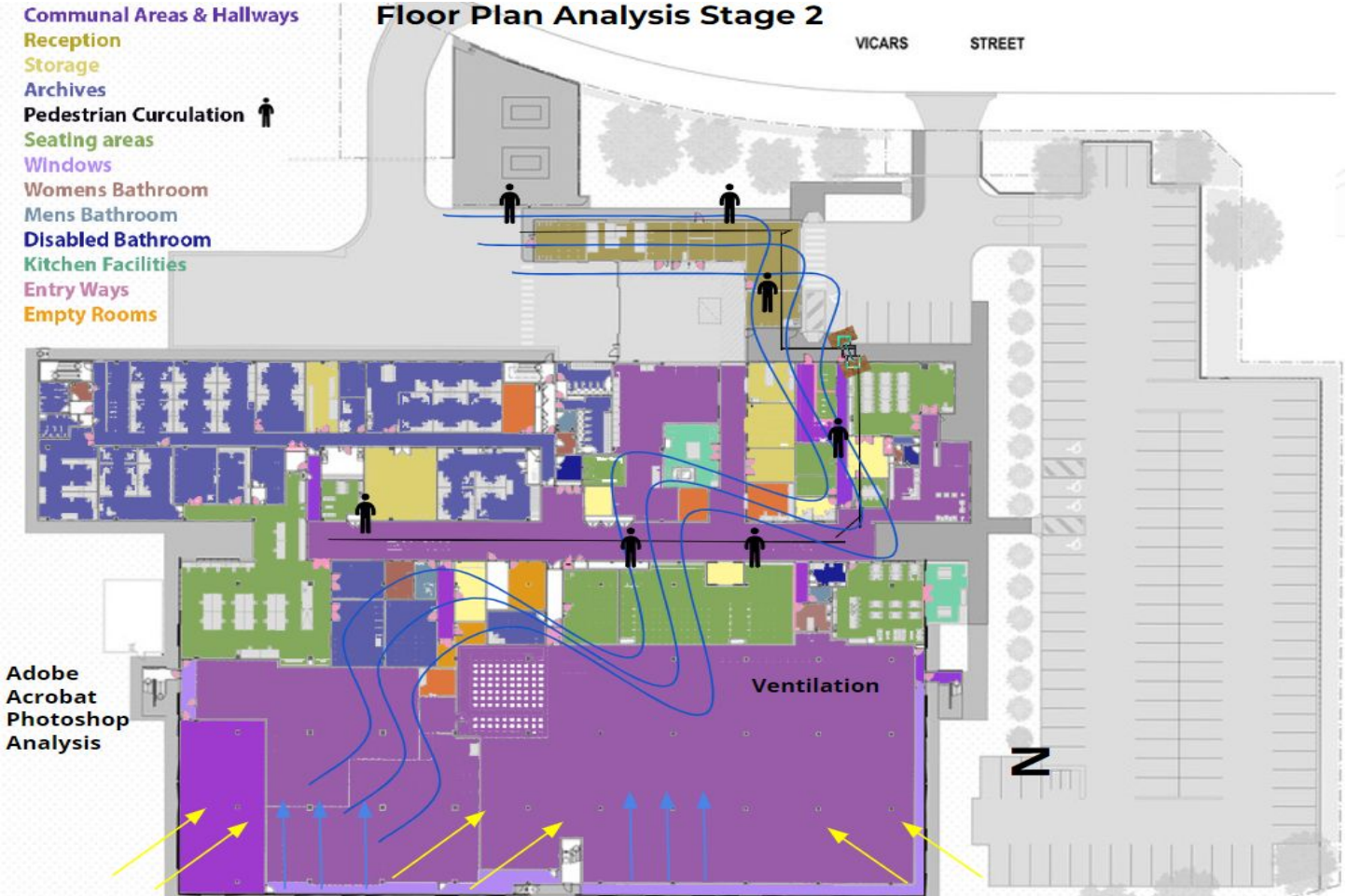
Analysis of Parking Facilities: The parking options are available next to the facility, including on-site parking areas and nearby public parking lots and garages. There is a large capacity of space, the layout enables more e=than enough access into the building.

Road Access: The facility is accessible by road including freeways and local roads, with designated entrances and parking areas for visitors and staff. Roads leading to the facility may are well-marked and easily navigable, allowing for convenient access by car or other private vehicles.

Natural Light: The site benefits from abundant natural light throughout the day due to its open exposure and minimal obstruction from surrounding buildings or vegetation. During daylight hours, sunlight penetrates deeply into the site, creating well-lit outdoor spaces and reducing the need for artificial lighting.Although, as it is not obstructed by vegetation during the hottest part of the day, it may require other elements to the building to ensure that it does not get too hot within the building.



Floor Plan Analysis Stage 2



Floor Plan Analysis Stage 2 Continued.

Spatial Relationships: Administrative areas are conveniently situated near the entrance for easy accessibility by employees and visitors. Conversely, storage and loading zones are located towards the rear or side of the building, minimizing disruption to office operations.

Natural Light and Views: Incorporating large windows or skylights into the design introduces natural light into the warehouse space, reducing reliance on artificial lighting and enhancing the overall workspace ambiance.

Furniture Placement: Office furniture within the administrative areas of this warehouse are thoughtfully arranged to maximize space utilization and functionality. Desks, chairs, and filing cabinets are strategically positioned to facilitate workflow and foster effective communication

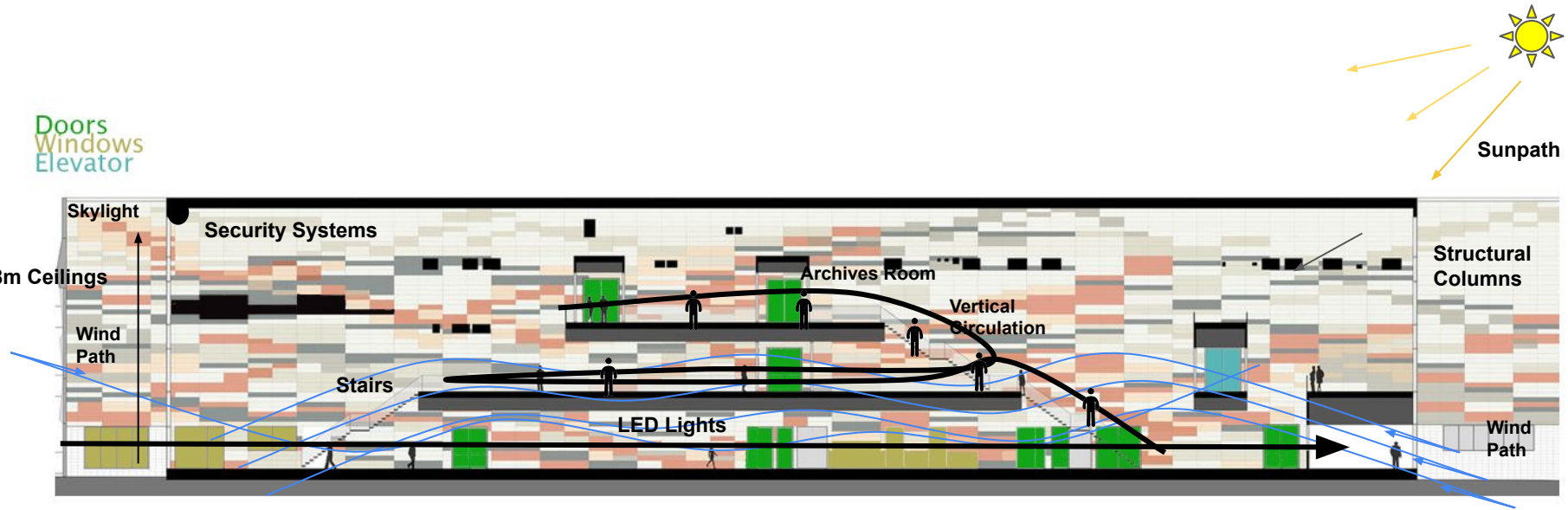
Room Layout and Functionality: Within this warehouse, the primary floor area is designated for storage and archive inventory management, featuring expansive, open space with high ceilings tailored for efficiently storing large quantities of goods and materials.

Circulation: The layout of this warehouse prioritizes efficient circulation pathways to facilitate the smooth movement of people and goods throughout the space. Wide aisles and unobstructed pathways are meticulously designed to accommodate the passage of forklifts, pallet jacks, and other equipment essential for warehouse operations. Additionally, strategic placement of loading docks and access points ensures seamless ingress and egress of goods, minimizing congestion and optimizing workflow. By prioritizing circulation in the floor plan design, the warehouse promotes productivity, safety, and operational efficiency within its dynamic environment.

Wind and Sun Path Analysis: The warehouse-style building is designed with sustainability in mind, incorporating an analysis of wind patterns and sun paths. By strategically orienting the building and maximizing natural ventilation and daylight, it minimizes energy consumption and promotes occupant comfort. This approach reduces reliance on artificial lighting and cooling systems, aligning with principles of environmentally conscious architecture.

Ventilation Optimization: The warehouse-style building prioritizes ventilation to enhance indoor air quality and occupant comfort while minimizing energy consumption. Through strategic placement of windows, vents, and airflow pathways, the design facilitates natural ventilation, reducing the need for mechanical systems and promoting energy efficiency. Additionally, the layout and orientation of the building is optimised to capture prevailing winds and promote airflow throughout the space, further enhancing ventilation effectiveness. By integrating ventilation optimization into the design process, the warehouse fosters a healthy and sustainable indoor environment conducive to productivity and well-being.

Vertical Site Analysis Stage 3



Vertical Analysis Continued.

Ceiling Heights: Within the National Archives preservation facility, the central storage area boasts generous ceiling heights, averaging at 18 meters. This ample vertical space accommodates standard archival shelving units efficiently. Additionally, specialized sections designated for oversized archival materials feature even higher ceilings, reaching up to 25 meters, ensuring optimal storage conditions for large artifacts and documents.

Structural Elements: Throughout the warehouse, strategically positioned structural columns ensure robust load-bearing capacity and structural integrity. These columns are meticulously placed to minimize obstruction to storage layout and circulation paths, thereby enhancing accessibility and operational efficiency.

Vertical Circulation: Vertical movement within the facility is facilitated by centrally located staircases and freight elevators. These elements are thoughtfully designed to streamline the movement of archival staff and materials between storage levels, promoting seamless workflow and expedited retrieval processes.

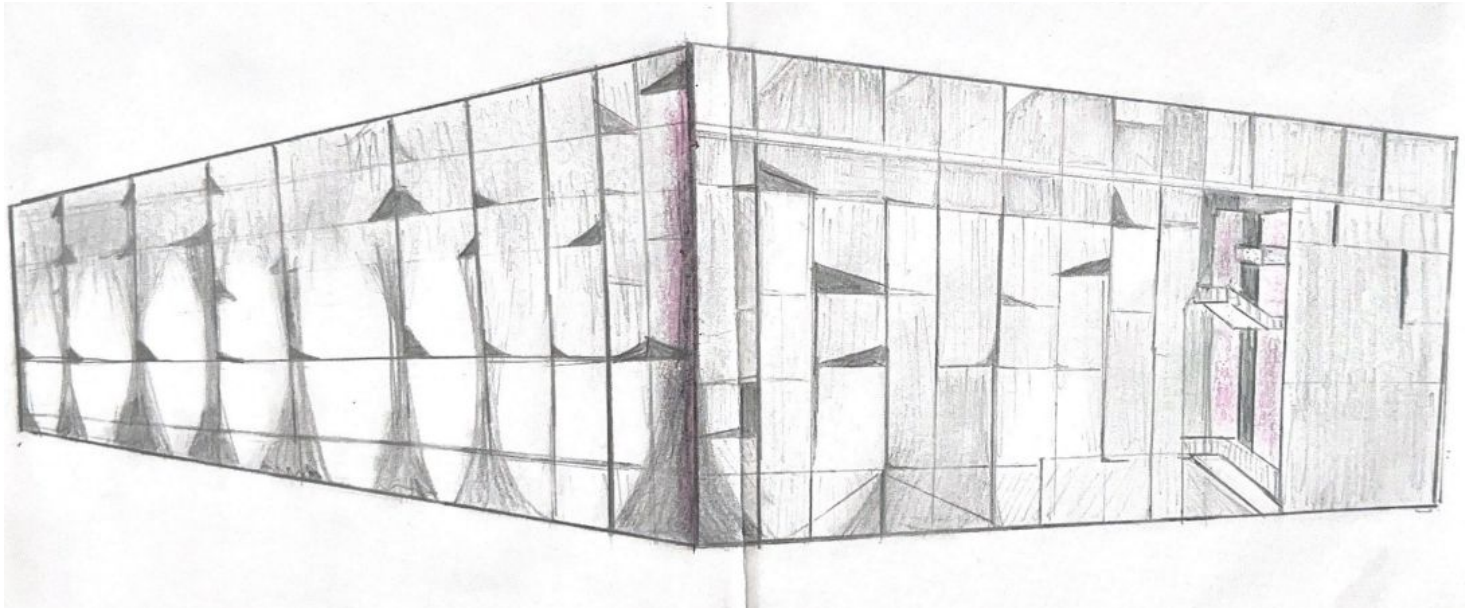
Spatial Zoning: The warehouse is meticulously divided into specialized zones tailored to the unique requirements of archival materials. Climate-controlled areas provide optimal preservation conditions for fragile documents, while strategically arranged high-density shelving systems maximize storage capacity and organizational efficiency.

Natural Light: Thoughtful integration of natural light sources, including strategically positioned windows and skylights, illuminates the warehouse space. Utilizing advanced light diffusers and shading systems, daylight levels are carefully regulated to protect archival materials from UV exposure while enhancing the overall ambiance of the facility.

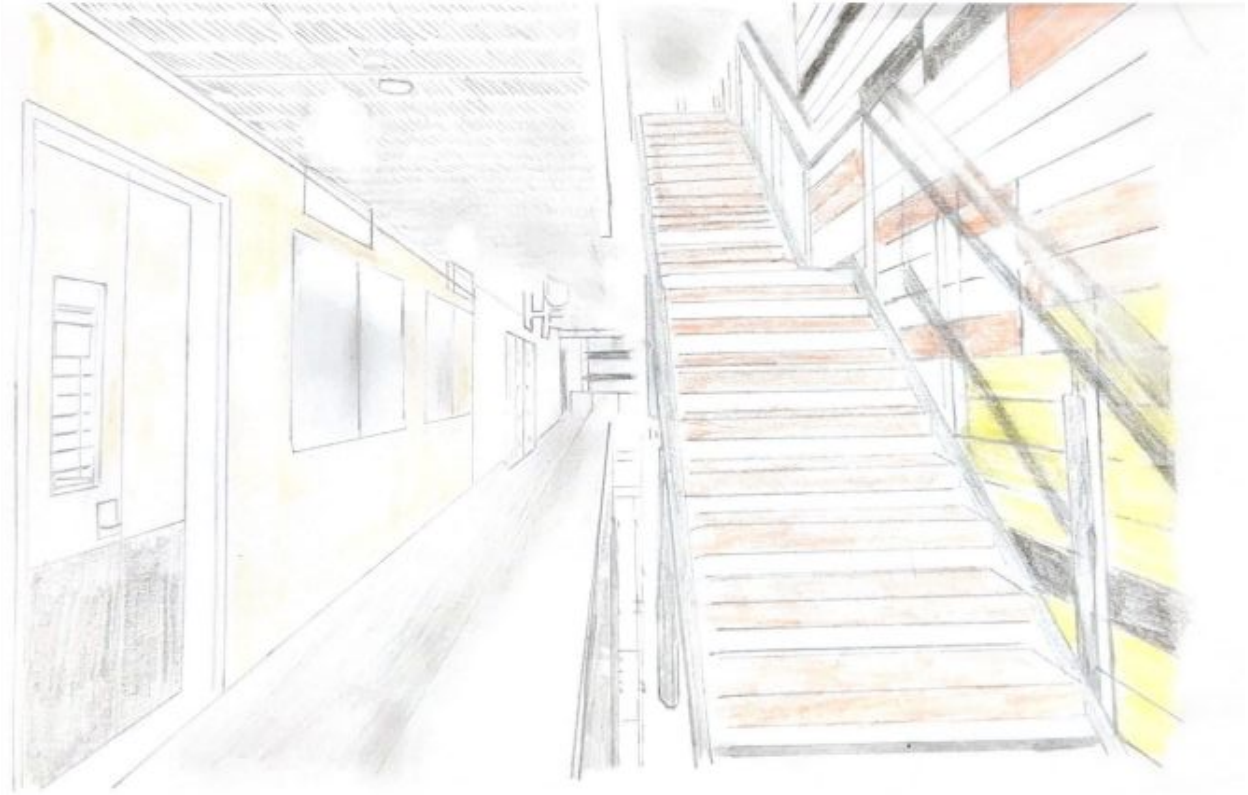
Artificial Lighting: Cutting-edge LED lighting fixtures are strategically deployed throughout the warehouse, offering customizable brightness and color temperature settings. Equipped with motion sensors and daylight harvesting controls, these fixtures optimize energy efficiency by dynamically adjusting lighting levels based on occupancy and natural light conditions.

Security Measures: Robust security measures are implemented to safeguard archival collections against unauthorized access and potential threats. State-of-the-art surveillance systems, access control protocols, and fire detection systems provide comprehensive protection, ensuring the integrity and security of valuable archival materials.

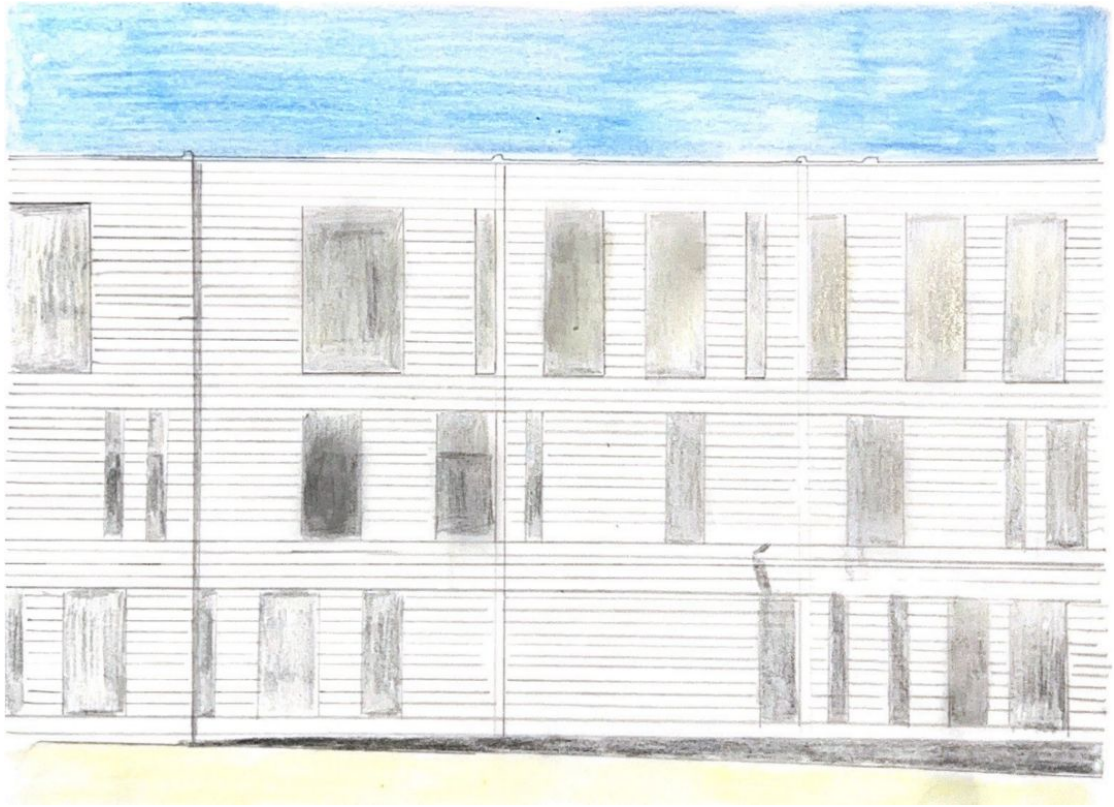
Hand Sketching Stage 1



Hand Sketching Stage 2



Hand Sketching Stage 3





Plans and Timeline for my model of the National archives Preservation

Shopping List:

- Foam Board (A1 5mm) $9.54
- CardBoard (3mm 900x600mm) $4.96
- Clear Contact Paper (5mm) $5.85
- Clear Gorilla Glue (51ml) $12.95
- Gorilla Emergency Repair Tape, 8.2 Meter Length, Clear $19.43
- Coloured Markers $2.00
- Quill A4 280gsm Kraft Board 25 Pack $31.29
- Quill A3 Flute Board Assorted 5 Pack $13.49
- Fiskars Softgrip Craft Knife $17.39

Utensils at home:

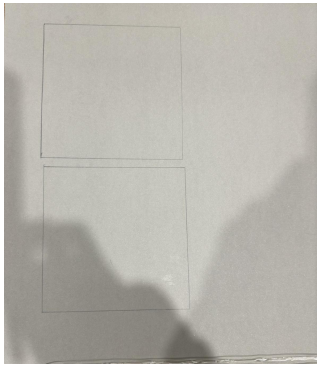
- Scissors
- Ruler
- Stencil
- Stanley Knife
- Sharpener
- Eraser
- Pencils

Plan:

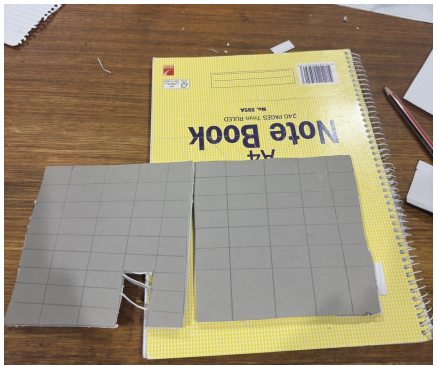
1. Utilizing printed drawings to trace and replicate shapes and structures onto foam board.
2. Employing kraft board as the foundation/base material for the model.
3. Crafting the model by shaping foam board into desired forms.
4. Colour in the cut outs to replicated colour pallet.
5. Adhering primary structural components together using glue.
6. Incorporating clear contact paper sheeting for windows.
7. Utilizing silver flute board to represent metal roofing materials
8. Allocate additional days for the adhesive to fully dry.
9. Completion of Model.

Time Management	Date	Completion Status
Materials	31/03/2024	✅
Started Marking out measurements on Foam Board	31/03/2024	✅
Cutting out parts of model	31/03/2024	✅
Assembling Model	02/04/2024	✅
Final touches and colour selections	04/04/2024	✅
Finish Project	12/04/2024	✅

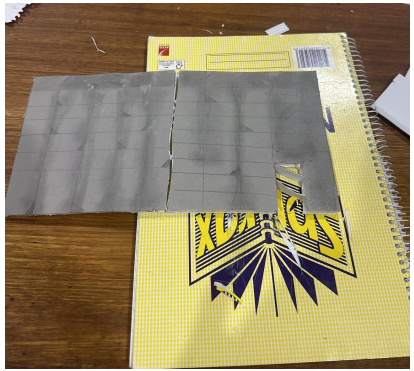
Physical Model Progress



Transferring the outlines and dimensions of wall and concrete panels onto foam board.

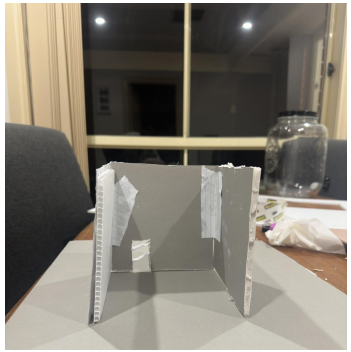


Indicating and marking out the areas where dimensions will be shaded and constructing the side access stairway, making sure they fit in the opening created.

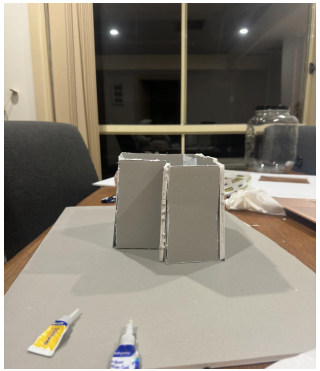


Rendering the facade to replicate the textured appearance depicted in sketches and reference photos.

Physical Model Progress Continued.



Affixing the concrete panels together with super glue and connector paper to guarantee their stability and cohesion.
As depicted above, showcasing the final outcome with securely assembled walls, exemplifying the successful application of super glue and connector paper to ensure structural integrity.

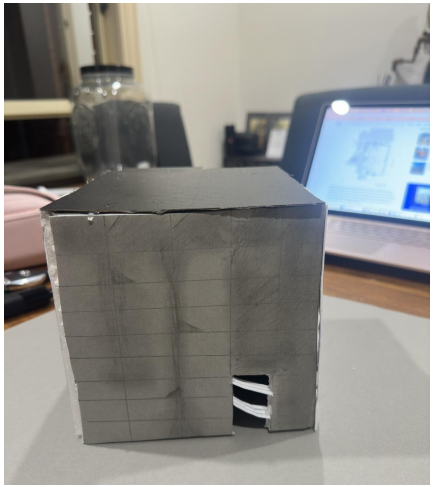




Physical Model Progress Continued.



Constructing the rear window wall using plastic material to serve as the windows, and exhibiting the precise cutouts for the windows as illustrated in the image above.

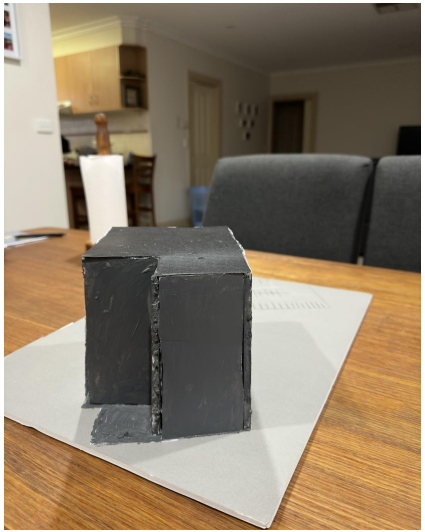


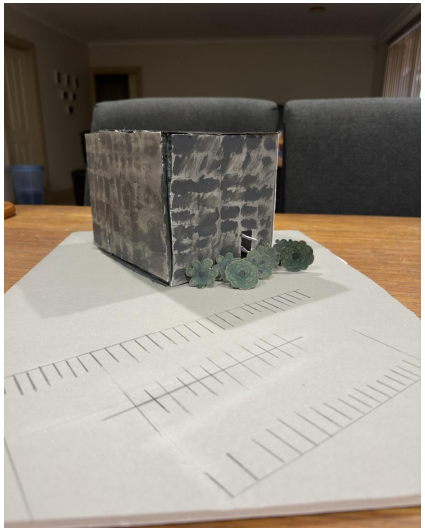
Evaluating the roof to verify the accuracy of measurements for the roofing covering over the entirety of the building.

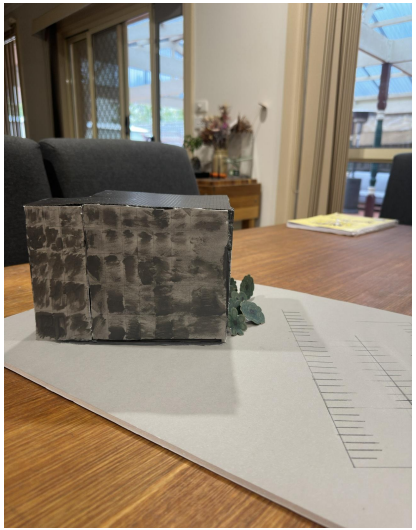


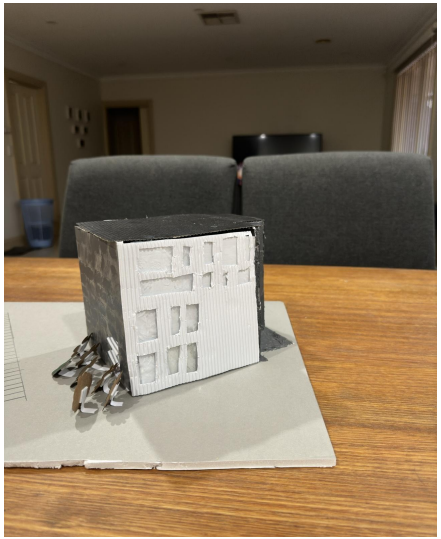
Upon the completion of securing the roof, the next step entails allowing it to dry thoroughly. In the ensuing days, I will commence adding final touches and accents to the project.

Final Model









Scale 1:100

Reflection and Plan for my Model Stage 1

As I delve into constructing my model, I've come to realize that achieving precision and authenticity demands thorough groundwork and unwavering commitment to each phase of the endeavor. The journey commences with extensive research, delving into architectural blueprints and reference materials to grasp the intricate dimensions and characteristics of the structure, ensuring meticulous adherence to scale. This foundational understanding informs my material selection, ensuring alignment with the intended scale and level of intricacy. Armed with precise tools like rulers and cutting implements, I meticulously measure and shape each component, striving for immaculate lines and exact angles. The assembly process unfolds methodically, commencing with the groundwork and steadily layering components while regularly consulting my research to uphold fidelity. Patience becomes paramount, allowing adequate drying time for adhesives and reinforcing structural integrity as needed. Throughout this journey, I continuously reference my research, making adjustments to maintain faithful representation. With each stroke and detail, I endeavor to encapsulate the essence of the structure, transforming it into a tangible manifestation of my creative vision. Through this hands-on experience and unwavering dedication to authenticity, my model transcends mere craftsmanship—it becomes a testament to my passion for architectural precision and meticulous detail, all carefully rendered to scale.

Short Reflection/ Graphic Style Plan Stage 1

Choosing Global Industrial Architecture was the focus of our project, a decision fueled by a blend of personal interests and academic aspirations within our group (Maisie, Ornela, Sam and Noor).

In our collaborative efforts, we consciously opted for foam board as the primary material for our model, driven by a collective appreciation for its minimalist aesthetics and its ability to deliver a refined finish. Recognizing the importance of clarity and simplicity in architectural representation, we found that foam board offers the ideal canvas to accentuate the core elements of our design without unnecessary embellishments. Its adaptable nature allows us to effortlessly sculpt and mold it to mirror the intricate details and contours of our vision, ensuring an accurate and visually striking portrayal. By selecting foam board, we ensure that our model will exude a sense of coherence and sophistication, effectively capturing the essence of our design with precision and finesse.

Industrial architecture, with its fusion of functionality, innovation, and historical relevance, offers a captivating narrative of human creativity and progress. It transcends mere structures, delving into the narratives of societal needs, technological advancements, and the dynamic evolution of landscapes. This exploration not only enriches our understanding of architectural evolution but also serves as a mirror reflecting global trends and challenges in construction and engineering. Each industrial site, influenced by its unique cultural, environmental, and economic context, offers a rich tapestry of architectural styles, construction methods, and materials to study and appreciate.

Moreover, our interest in industrial architecture extends beyond its aesthetic and historical value. It presents an intriguing canvas for exploring sustainability and adaptive reuse a subject increasingly pertinent in contemporary construction discourse. By examining successful adaptive reuse projects, we aim to glean insights into repurposing existing structures, minimizing waste, and fostering resilient, environmentally conscious built environments.

Individually, our fascination with industrial architecture varies, reflecting diverse interests within the realm of construction. Some of us are drawn to the historical narratives embedded in industrial sites, while others are captivated by the potential of sustainable design principles or the integration of technology in modern industrial facilities. This project provides an avenue for each of us to pursue our unique passions while contributing to a collective endeavor.

Selecting the National Archives Preservation Facility as my individual case study within our group's broader exploration of industrial architecture around the world was a decision rooted in a blend of personal fascination and strategic alignment with our overarching project goals.

On a personal level, the National Archives Preservation Facility captivates me due to its unique intersection of history, architecture, and function. It's not just a repository of documents; it's a custodian of our collective memory, safeguarding the records that tell the story of our nation's past. The facility's architectural design, with its emphasis on preservation and symbolism, speaks volumes about the importance of heritage conservation in a rapidly changing world.

Moreover, choosing the National Archives Preservation Facility aligns seamlessly with our group's larger mission of exploring industrial architecture on a global scale. While it may seem distinct from traditional industrial structures like factories or warehouses, the facility embodies the principles of industrial architecture in its design, construction, and purpose. It showcases how architectural innovation and engineering prowess can converge to create a functional yet aesthetically compelling industrial space dedicated to the preservation of cultural heritage.

Group Topic Theme & Further Reflection Stage 2

Our exploration of industrial architecture across different regions highlighted the diversity of design approaches, construction techniques, and historical contexts that shape built environments worldwide. Each case study, including my analysis of the National Archives Preservation Facility, provided a unique lens through which to examine the evolution of industrial structures and their impact on society.

My journey commenced with a meticulous examination of the site surrounding the National Archives Preservation Facility. Through site analysis, I discerned not only the physical characteristics but also the historical and cultural context that informed the facility's design and location. By scrutinizing factors such as topography, orientation, and surrounding infrastructure, I gained invaluable insights into how the site's features influenced the architectural decisions and functionality of the facility.

Moving further, my floor plan analysis delved into the intricate details of the floor plan, unraveling the spatial layout and functional organization of the facility. With a keen eye for detail, I dissected each room and space, deciphering their intended purposes and interrelationships. The floor plan served as a blueprint for understanding circulation patterns, access points, and the integration of structural elements, guiding me in comprehending how users navigate the space and interact with its features.

Beyond the physical attributes, my analysis ventured into the realm of architectural symbolism and cultural significance. By scrutinizing the floor plan and architectural elements through a lens of cultural context, I uncovered layers of meaning embedded within the design. From the arrangement of spaces to the choice of materials, every aspect of the facility's architecture resonated with historical narratives and cultural values, reflecting a profound connection to the national identity it seeks to preserve.

Moreover, my analysis extended beyond static representations to encompass dynamic considerations such as sustainability and adaptive reuse. By evaluating the facility's environmental impact and potential for future adaptation, I gained a holistic understanding of its long-term viability and relevance in a rapidly changing world.

In essence, my reflection on the analysis of the National Archives Preservation Facility transcended mere observation, evolving into a nuanced exploration of its historical, cultural, and functional dimensions. Through the synthesis of site analysis, floor plan examination, and cultural contextualization, I uncovered the intricate interplay of factors that shape the built environment and its enduring impact on society.

This journey has been incredibly enlightening, allowing me to contextualize my individual case study within the broader framework of global industrial architecture. It has underscored the importance of understanding the historical, cultural, and functional aspects of industrial structures and their enduring impact on the built environment. Through this collective exploration, I've deepened my understanding of architectural heritage and its significance in shaping the world we inhabit.

Group Topic Theme & Further Reflection Stage 2 Continued.

Exploring the National Archives Preservation Facility has truly broadened my understanding of architecture's role in preserving history and embracing modernity. Each aspect of the facility's design reflects a careful consideration of the past, present, and future.

The precast concrete form, reminiscent of Australia's rugged landscape, serves as a tangible connection to our heritage while prioritizing functionality through thermal efficiency. It's a testament to the enduring importance of safeguarding our cultural legacy.

Conversely, the space designated for archivists embodies a forward-thinking approach, utilizing lightweight materials and modular design principles to accommodate the evolving needs of archival work. This juxtaposition between tradition and innovation highlights the facility's adaptability and readiness for future challenges.

Archive Row, the central circulation spine, emerges as more than just a physical connector; it symbolizes collaboration and unity among those tasked with preserving history. It fosters an environment where archivists can come together, share knowledge, and celebrate their vital role in society.

The light-filled atrium, with its clay panels depicting the transition towards digitized records, serves as a visual representation of the evolving nature of archival practices. It prompts reflection on the intersection of tradition and progress, reminding us of the importance of embracing technological advancements while honoring our past.

In essence, the National Archives Preservation Facility embodies the essence of global industrial architecture by seamlessly blending tradition with innovation. It stands as a testament to the enduring significance of preserving our cultural heritage in an ever-changing world.

Group Topic Theme and Further Reflection Stage 3

Through the extensive exploration of the National Archives preservation facility within the context of global industrial architecture, I've acquired invaluable insights into the intricate interplay among historical preservation, architectural design, and industrial functionality. This journey has been both enlightening and challenging, deepening my understanding of how built environments intersect with cultural significance worldwide.

Examining the National Archives preservation facility has emphasized the critical significance of adaptive reuse and sustainable design principles in revitalizing industrial spaces for modern use. From scrutinizing the facility's vertical section to analyzing its spatial layout and environmental considerations, I've come to recognize the multifaceted approach necessary to balance preservation goals, functionality requirements, and user needs.

Furthermore, delving into the broader context of global industrial architecture has expanded my perspective on the diverse architectural styles and construction methods employed globally. This study has underscored the pivotal role of industrial architecture as a tangible embodiment of cultural heritage and technological progress, shaping the identities of communities and nations around the world.

Ultimately, this reflective journey has underscored the importance of interdisciplinary collaboration and holistic problem-solving in tackling the intricate challenges of architectural preservation and industrial heritage. By integrating insights from various disciplines such as architecture, engineering, history, and cultural studies, innovative solutions can be developed that honor the past while embracing the future of global industrial architecture. As I move forward, I'm inspired to apply these newfound understandings and skills to contribute meaningfully to the preservation and advancement of our built environment on a global scale.

Group Theme

In line with our group's vision for simplicity and clarity, our presentation format adheres strictly to a black and white minimal aesthetic. Each slide and visual element has been meticulously curated to maintain a clean, uncluttered appearance, reflecting our commitment to effective communication and streamlined design. By embracing this minimalist approach, we aim to focus attention on the essential content of our presentation, ensuring that our message is conveyed with maximum impact and clarity.



Brewery Hall

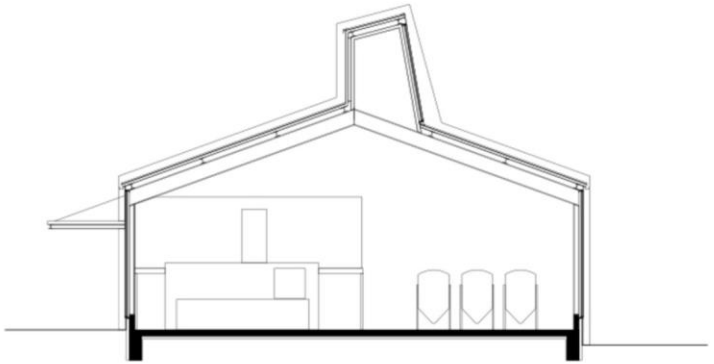
Made by Sam Ibrahim

Analysis

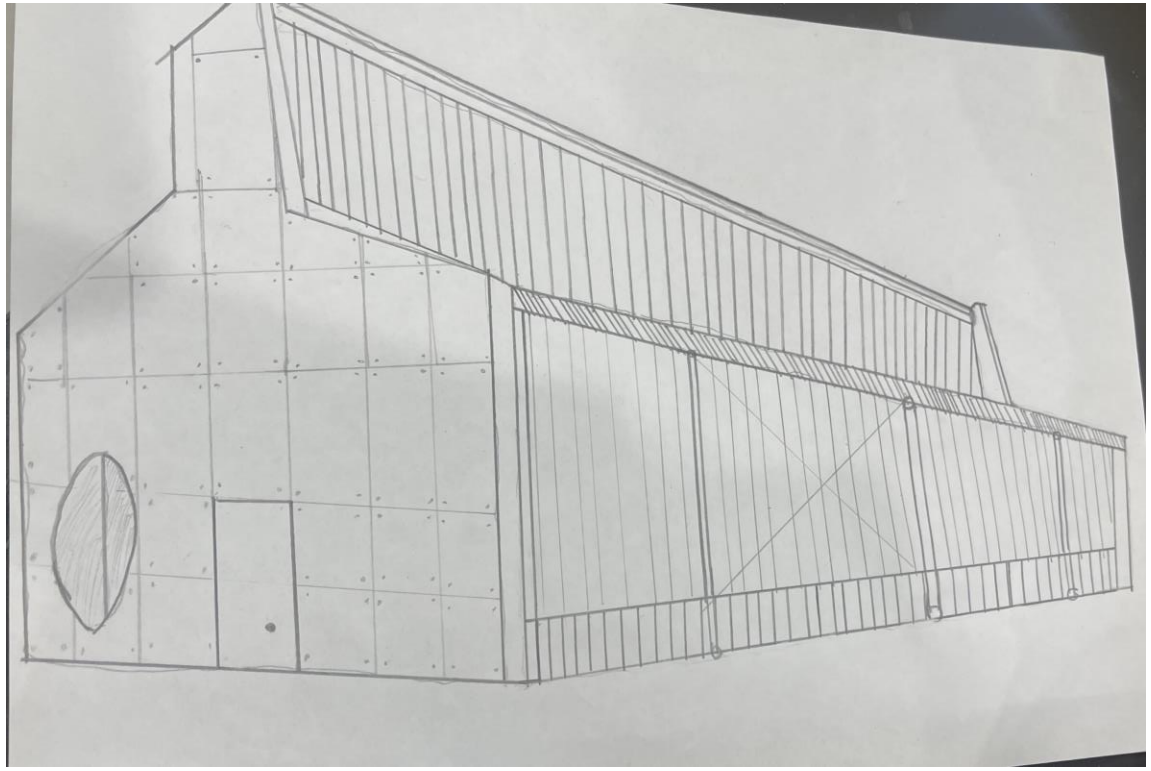
- Site Analysis:

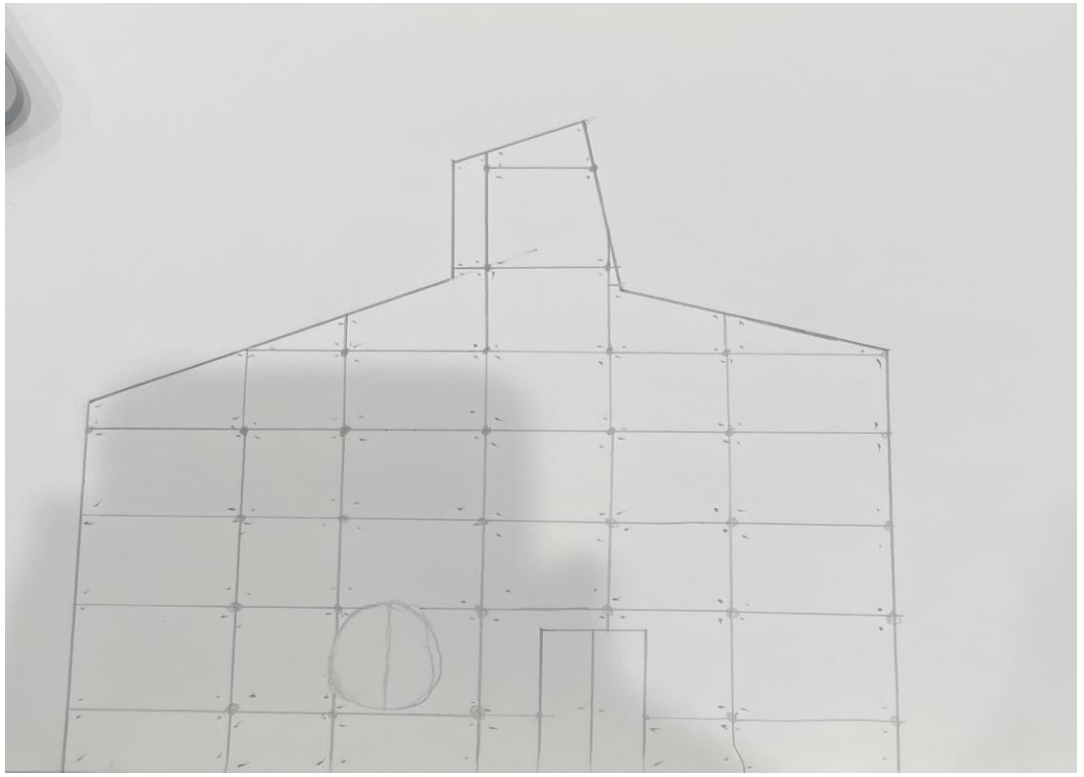


Floor analysis:



Perspective drawing





MATERIALS

- PAINT

Black paint, white paint, and silver paint.

Black paint to mix with white to make grey for the concrete. Silver paint was used to paint the roof

- FOAM BOARD

Foam board was used to place the model on and was painted as concrete for the floor

CARDBOARD

Cardboard was mainly used to build the model like the roof and sides of the building

- SCISSORS

Scissors were used to cut out the cardboard to make the model

- POPSICLE STICKS

The sticks were glued together to make the wall of the building look more defined like the picture

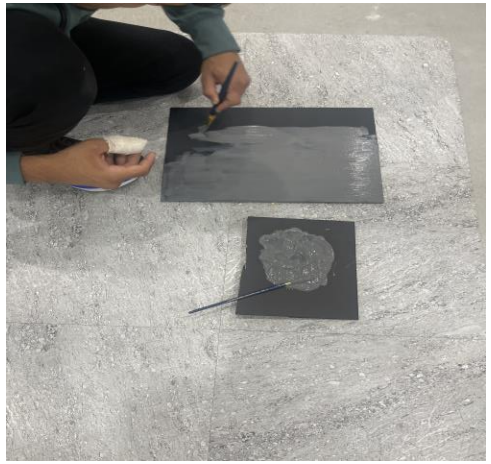
- GLUE GUN

The glue gun was used to glue all the pieces of the model together

- PAINT BRUSHES

Paint brushes were used to paint the model depending on how thick the painting should be

Steps on building the model



Painting the foam board to create the concrete and let it dry



Cutting out the pieces of carboard to form the model and gluing them with the glue gun



Model is complete and just needs to be painted



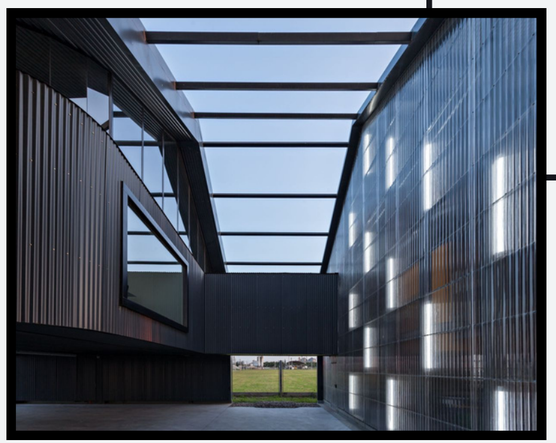


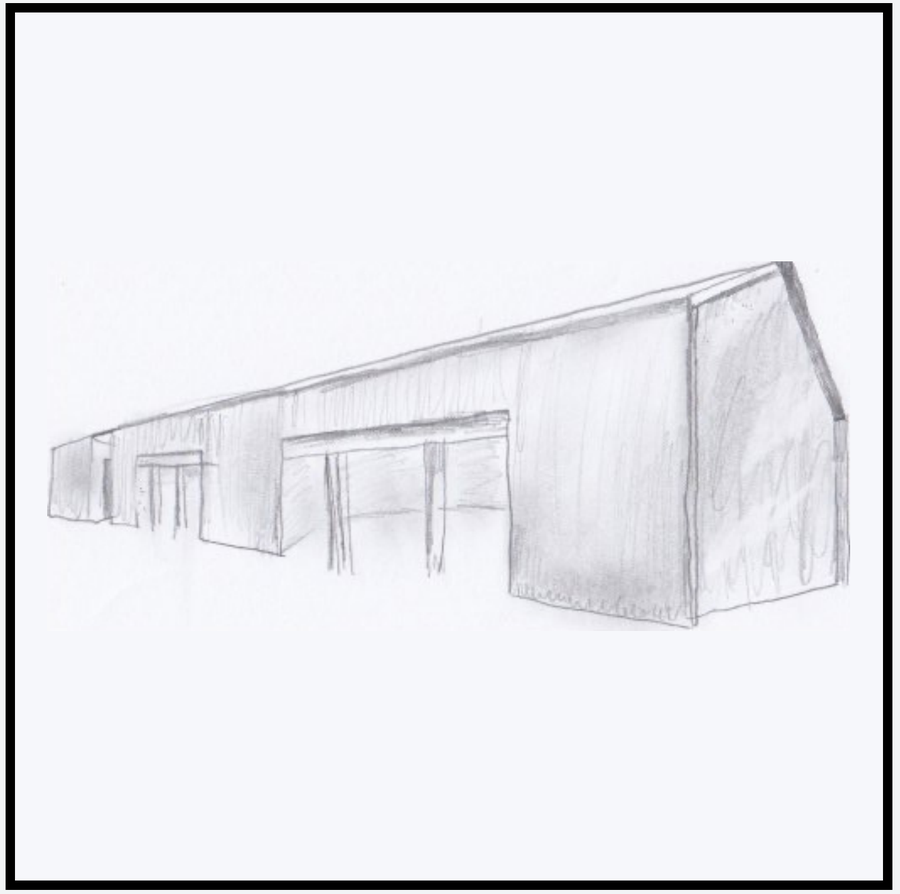
Model has been painted and completed. Added a road to match a little bit with the site plan



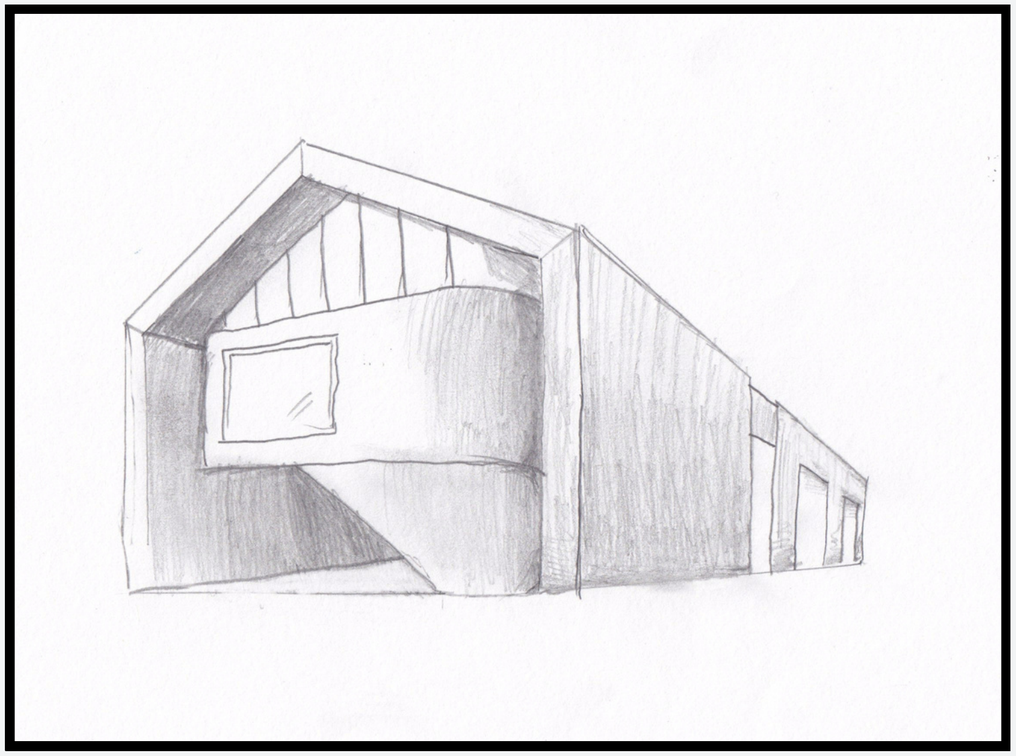
BERTOLOTTI VIAL WAREHOUSE AND OFFICE

SANTA FE, ARGENTINA

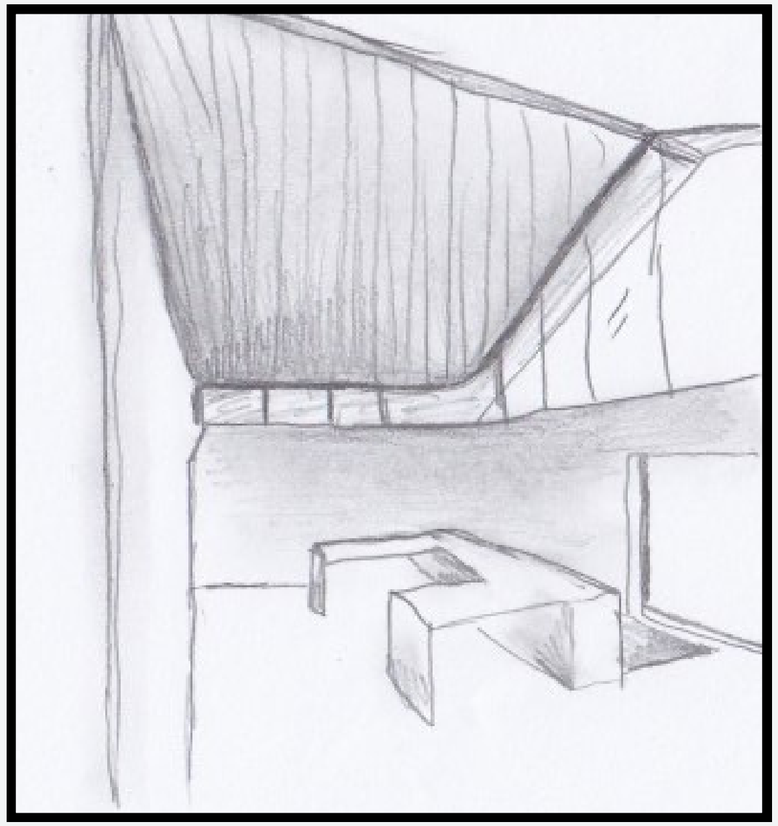




PERSPECTIVE DRAWINGS



PERSPECTIVE DRAWINGS



PERSPECTIVE DRAWINGS

PHYSICAL MODEL PLAN

Materials
Foamboard
Clear adhesive
Scissors/Stanley knife/box cutter
Any clear plastic material to build windows.
Cardboard for base of the model.

Step by Step Plan

1. I am planning on using the cardboard for the base where the model will sit.
2. Start cutting out the shapes of the model from the foam board. e.g, the walls, roofing, openings etc.
3. Once all the shapes have been cut, they need to be glued all together using the clear adhesive to create the shape of the model.
4. Then the spaces that I have left open for the windows I am going to cut out pieces of clear plastic to add the windows.



After everything is glued together and I have the basic shape of the model, I am also planning to add that ribbed like texture to the model so it resembles the building a bit more. Alongside that I want to also use foam board that has a similar colour.

SITE ANALYSIS

Site



- From the site analysis we can see that the barn is located in an industrial area in which these areas consist of many warehouses, factories, distribution centres, etc. We can see the same features in this site analysis as the barn (blue) is surrounded by other industrial buildings.
- Alongside, there is a lot of greenery surrounding it which could indicate that there was elements of sustainability implemented during the design process of this building.

Legend

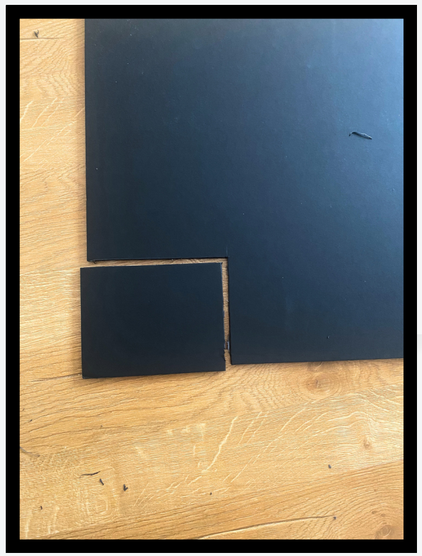
- Roads
- Greenery/Vegetation
- Other warehouses/barns
- Bertolotti warehouse

Analysis

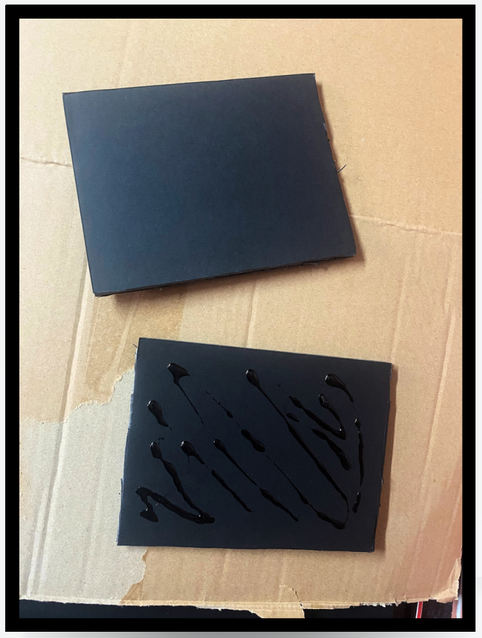


- Another prominent feature of this site is the roads which are going in straight directions without too many turns. This indicates the kind of vehicles that are driven on those roads which are typically bigger vehicles such as trucks.

PHYSICAL MODEL PROCESS



Cutting out the shapes from the foam board to make the walls.



I cut out two of every shape and stuck them together to make the walls thicker.



The side walls have openings for a garage so I cut the openings out.

This is what it looked like when I stuck the pieces together and onto the cardboard. Now all thats left is to add the roof, a wall at the back and a window at the front.



Scale: 1:100

PHYSICAL MODEL



FLOOR PLAN ANALYSIS

- This building is a very well ventilated building due to it containing a lot of openings and entries (as shown with the blue arrows).

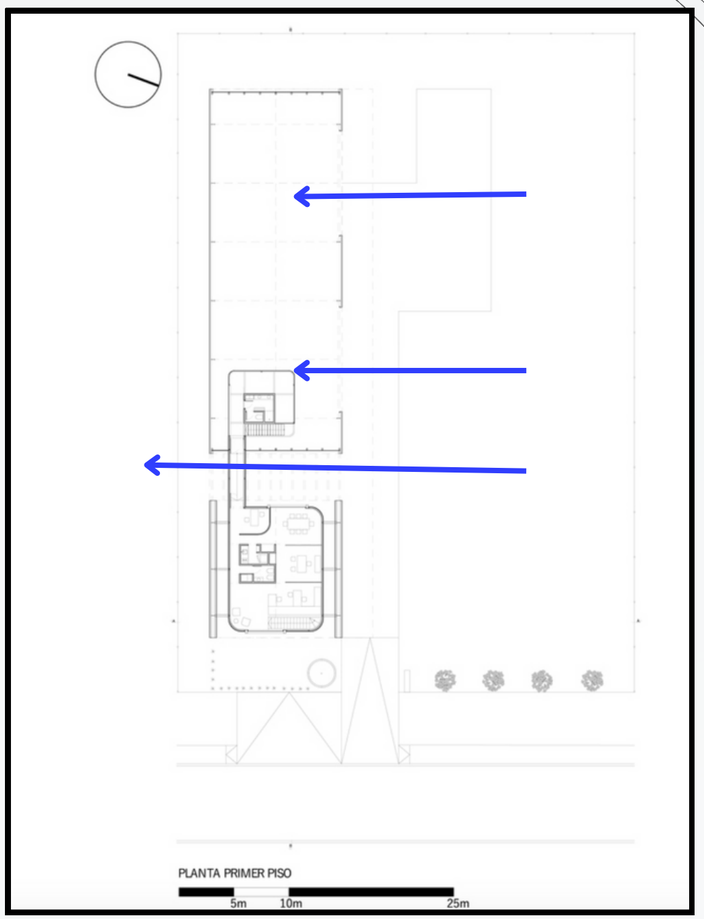
- The amount of opening allows for a lot of sunlight to flow throughout the whole building (as shown by the yellow arrows).

LEGEND

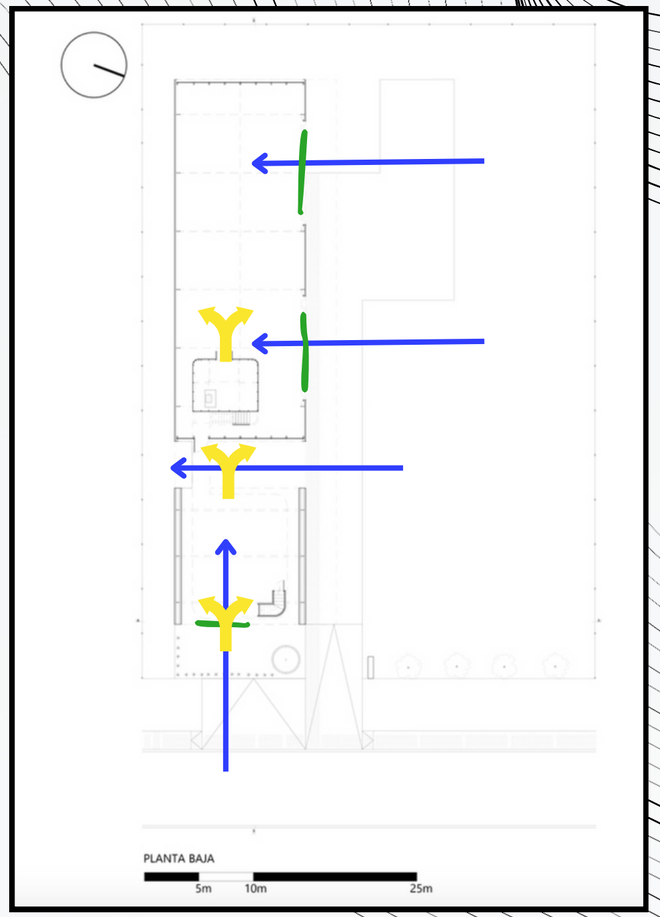
Ventilation

Sunlight

Entry/Openings



First floor



Ground floor

REFLECTION



As a group we decided to choose industrial architecture as our focus as it is an interesting aspect of the built environment, and its very ‘purpose driven’ as opposed to other architectures where the aesthetic and appearance is prioritised. Industrial architecture also shows us how far technology and industry have evolved over time, showing how adaptable these kind of buildings are to the current industrial and societal needs. Another aspect to be explored is how industrial architecture impacts urban development as these buildings are crucial parts of the urban ecosystem as they contribute to the economy and provide many employment opportunities.



My personal interest led me to focus specifically on the Bertolotti Vial Machinery and Office Warehouse which is also classified as a barn. What drew me to this building was the very minimalistic appearance it had compared to many other industrial buildings. Aside from the appearance I was interested in the spatial aspect of the building in terms of how space is optimised in a way where the workflow is smooth while also ensuring safety. I also wanted to look into how technology is implemented and how they could be impacting productivity and possibly reshaping the industry.

REFLECTION

Throughout creating the physical model I learned numerous things not only about industrial buildings but also from the process of creating a physical model. One of these things was how the use of different materials can impact the final outcome of the model depending on the colour, texture, or size etc. These can affect both the aesthetic and quality of the model. For my model specifically, I found that using foam-board gave my model a more minimalistic appearance due to its matte finish. I also gained an appreciation for the details that are implemented into industrial buildings as often times they can be very easily overlooked. I learned that the way in which different elements are included can create different kinds of spatial experiences specifically the openness of the building as that is one of it's key features. One key takeaway from this design was how both aesthetics and functionality are balanced when creating a design. The Bertolotti Vial Machinery and Office Warehouse integrates office spaces with machinery and warehouse functions which exemplifies how a building can still possess operational efficiency while also being visually appealing.

GROUP THEME

Topic: Industrial Architecture

- *Color Palette: Black, White & Grey*
- *Layout: Landscape*
- *Theme: Minimalistic*